

The One-Axion Universe: A Computational Framework for Simulating Benevolent Symmetry Restoration

Version 2.0 — Revised & Scientifically Deepened | June 2026

Classification: Tardigradia Game Engine | Physics Deep Dive Series | TGPU v2.0 **Particle Class:** Pseudoscalar Boson (BSM) | Spin: 0 | Charge: 0 | Mass: $\sim 10^{-5}$ eV (hypothetical) **Source:** Extends v1.0Axion Deep Dive Emulation.pdf (V1.0)

Δ Change Log V1.0 → V2.0

Parameter	V1.0	V2.0	Source
HAYSTAC quantum squeezing	Standard microwave cavity resonance search	HAYSTAC Phase II (2023): quantum squeezed vacuum states reduce noise below standard quantum limit by 15.1 dB — doubles effective scan rate for axion dark matter search	Backes et al., Nature 2021; HAYSTAC Phase II 2023
ORGAN millikelvin resonator	General haloscope concept	ORGAN (2023, Univ. Western Australia): first axion haloscope operating at 26 GHz (millikelvin cooled), targeting higher-mass axion parameter space ($m_a \sim 110 \mu\text{eV}$)	McAllister et al., ORGAN 2023
Axion dark matter window	General QCD axion window	V2.0 updated window: $m_a = 2\text{-}20 \mu\text{eV}$ (DFSZ model) or $20\text{-}200 \mu\text{eV}$ (KSVZ) — experiment reaching sensitivity to KSVZ coupling. CMB constraints: $m_a > 0.01$ eV if thermal; dark matter if $m_a < 0.01$ eV	Di Luzio et al., Physics Reports 2020 updated 2024
Fuzzy dark matter regime	Scalar field cosmology concept	V2.0: FDM axion ($m_a \sim 10^{-22}$ eV) creates quantum pressure at galactic scales — de Broglie wavelength $\sim$ kpc. Excludes cores of dwarf galaxies from JWST morphology constraints (2024)	JWST galaxy morphology 2023-2024; Rogers & Peiris 2021

Parameter	V1.0	V2.0	Source
CASPER-electric	Not covered	V2.0: CASPER-electric searches for axion-nucleon coupling via NMR-like effect. Mass range: 10^{-12} to 10^{-6} eV. 2024 result: first sensitivity to QCD axion in m_a region below 10^{-8} eV	CASPER Collaboration 2024
Bose-Einstein condensate simulation	Conceptual BEC reference	V2.0: Axion BEC coherence length $\xi = \hbar/(m_a * v_{rms})$. In simulation: when axion field density exceeds critical density ρ_c , switch to coherent field mode (wave) from particle mode. Phase transition rendered as ‘crystallization’ visual	Hui et al., Phys. Rev. D 2021

1. The One-Particle Universe Framework — V2.0 Update

The foundational architecture established in V1.0 — Worldline Monism, the “Origami Universe” metaphor, and the Object-Oriented ontology distinguishing intrinsic from extrinsic properties — is **fully preserved** and remains physically rigorous. V2.0 layers NEW discoveries from 2022-2026 atop this framework.

1.1 WebGPU Compute Architecture (V2.0 Core Infrastructure)

V1.0 designed for WebGL/Three.js on the CPU-side particle update loop. V2.0 upgrades to WebGPU (available in Chrome 113+ May 2023, Firefox 2024):

```
// WGSL Compute Shader - V2.0 Particle Integration
@binding(0) @group(0) var<storage, read_write> particles: array<ParticleState>;

struct ParticleState {
    position: vec4f,
    momentum: vec4f,
    spin: vec2f,           // Spinor components
    berry_phase: f32,      // Geometric phase accumulated
    proper_time: f32,      // tau - worldline parameter
    color_charge: u32,      // Packed color in 2 bits (R=0,G=1,B=2)
    quantum_numbers: u32,  // Packed: isospin, strangeness, charm, beauty
    decay_lifetime: f32,
    entanglement_id: i32,  // -1 = unentangled; >=0 = entanglement group
}
```

```

@compute @workgroup_size(64)
fn updateParticles(@builtin(global_invocation_id) id: vec3u) {
    let i = id.x;
    if (i >= arrayLength(&particles)) { return; }

    var p = particles[i];

    // Feynman-Schwinger proper time evolution
    let dtau = uniforms.dt / p.mass; // proper time step

    // Update worldline parameter
    p.proper_time += dtau;

    // Berry phase accumulation (topological phase)
    p.berry_phase += dot(uniforms.gauge_field, p.momentum.xyz) * dtau;

    // Lorentz force (for charged particles)
    let F = cross(uniforms.B_field, p.momentum.xyz) * p.charge_sign;
    p.momentum.xyz += F * uniforms.dt;
    p.position.xyz += p.momentum.xyz * uniforms.dt / p.energy;

    particles[i] = p;
}

```

1.2 Updated Physical Constants (CODATA 2022)

Constant	V1.0 Value	V2.0 Value (CODATA 2022)
m _e	9.10938356×10 ⁻³¹ kg	9.1093837139×10 ⁻³¹ kg
e	1.60217663×10 ⁻¹⁹ C	1.602176634×10 ⁻¹⁹ C (exact)
ħ	1.054571817×10 ⁻³⁴ J·s	1.054571817×10 ⁻³⁴ J·s (exact)
c	2.99792458×10 ⁸ m/s	2.99792458×10 ⁸ m/s (exact)
α	1/137.036	1/137.035999084 ±0.000000021

2. V2.0 Enhanced Data Structures

Extending V1.0 struct definitions with 2022-2026 discoveries:

```

// V2.0 Universal Particle State – extends V1.0
struct ParticleInstanceV2 {
    // Inherited from V1.0
    double proper_time; // tau worldline parameter (meter)
    glm::dvec3 position; // 3-space vector
    glm::dvec3 momentum; // 3-momentum (relativistic)
    float spin_z; // +0.5 or -0.5 (or eigenvalue)
    float berry_phase; // Accumulated Berry/Aharonov-Bohm phase

    // V2.0 ADDITIONS
    float entanglement_r; // Reduced density matrix purity 0-1
    int entanglement_partner_id; // -1 = none; global particle ID
    float flavor_oscillation_phase; // For neutrinos/kaons
}

```

```

uint32_t color_bits;           // 3-bit: R(b0),G(b1),B(b2)
float decay_probability_dt;    // P(decay in dt) for unstable particles
float vacuum_alignment;        // Order parameter for condensate mode

// WebGPU Buffer packing
// Total: ~100 bytes/particle * 10^6 particles = 100 MB GPU memory
float _pad[2];                 // Align to 16 bytes for WGS
};

```

6. V2.0 — Quantum-Enhanced Detection Architecture

The 2023 HAYSTAC result is the first axion search to operate below the Standard Quantum Limit (SQL) using squeezed vacuum states. This directly maps to a new simulation mechanic:

Quantum squeezing in simulation:

```

struct AxionDetector {
    // HAYSTAC-inspired quantum sensor model
    double standard_quantum_limit; // SNR = 1 at this axion coupling
    double squeezing_factor;        // V2.0: can be < 1 (sub-SQL)
    double effective_sensitivity;   // = SQL / squeezing_factor

    // Squeeze parameter r (measured in HAYSTAC Phase II: 15.1 dB => r = 1.74)
    static constexpr double HAYSTAC_SQUEEZE_dB = 15.1;
    static constexpr double SQUEEZE_r = 1.74f;
};

```

Axion field simulation update loop:

```

// V2.0 BEC phase transition detection
function updateAxionField(axionParticles, dt) {
    const meanDensity = computeMeanDensity(axionParticles);
    const criticalDensity = 1.878e-26 * h_param * h_param; // kg/m3

    if (meanDensity > criticalDensity * 0.85) {
        // Approaching BEC: switch to coherent wave mode
        transitionToCoherentMode(axionParticles);
        logEvent('[AXION] BEC threshold approaching. Coherence length: ' +
            coherenceLength(meanDensity).toFixed(2) + ' kpc');
    }
}

```

8. V2.0 Benevolence Metric Update

The “Benevolence” concept central to all OEU documents is given a precise V2.0 mathematical definition grounded in 2022-2026 quantum information theory:

Benevolence = Quantum Coherence + Information Preservation + Symmetry Maintenance

$$B(t) = C(t) \cdot F(t) \cdot S(t)$$

Where: - C(t) = l₁-norm coherence of particle density matrix (Baumgratz et al. 2014) - F(t) = quantum Fisher information / classical Fisher information (ratio ≥ 1) - S(t) = fraction of symmetries unbroken at time t

V2.0 game mechanic: Benevolence score increases when: - Particles maintain long entanglement coherence (high C) - Topological protections are active (Berry phase winding conserved) - Symmetry groups are preserved in interactions (no spurious symmetry breaking)

End of v1.0Axion Deep Dive Emulation V2.0 | Tardigradia Game Engine SubParticles Series This document supersedes V1.0 in all physics specifications.